

Teaching Manual: Place Value Explorer

Week 1, Day 1 | Scripted validation lesson | 60-75 minutes

[START HERE](#)

Lesson goal: the student understands that a digit's value is determined by its position in a base-ten system.

What the Student Should Learn

- Read and describe four-digit numbers using thousands, hundreds, tens, and ones.
- Connect a concrete model, a place-value chart, standard form, expanded form, and word form.
- Understand that 10 units in one place can be traded for 1 unit in the place to the left.
- Explain that moving a digit one place left makes its value 10 times as great.

Success Criteria

- The student says, for example, 'The 2 in 3,245 is worth 200,' not merely 'It is in the hundreds place.'
- The student treats zero as a placeholder in numbers such as 4,070 and 6,308.
- The student can trade 1 thousand for 10 hundreds without believing the total changed.
- The student can represent one number in at least three forms.

Important: Do not read every script word-for-word if the child is already reasoning well. The script gives the intended language and decision points. Pause often and let the student do the mathematical talking.

Lesson at a Glance

10 min	Diagnostic: number sense and mental math	Student pp. 1-2
7 min	Number talk: make 1,000	Loose pieces
15 min	Model 3,245 and connect representations	Mat + pieces
12 min	Zero placeholders and trading	Mat + digit cards
15 min	Guided and independent practice	Student pp. 4-6
10 min	Challenge and exit ticket	Student pp. 7-8

The Mathematics and Teaching Theory

What the teacher needs to understand before teaching

THEORY

1. Place Value Is Multiplicative

Our number system is base ten. Ten ones compose one ten, ten tens compose one hundred, and ten hundreds compose one thousand. A digit therefore has two meanings: its face value and the value created by its position. In 3,245, the digit 2 has face value 2 but place value 200.

Teacher implication: Always ask both 'What digit do you see?' and 'What is that digit worth?'

2. Concrete -> Pictorial -> Abstract

The student first handles or sees quantities, then organizes them on a place-value mat, and only then records symbols. The materials are not decoration: they make the ten-to-one relationship visible.

- Concrete: blocks, disks, or cut-apart value pieces.
- Pictorial: drawings or labeled columns on the place-value mat.
- Abstract: 3,245; $3,000 + 200 + 40 + 5$; number words.

3. Variation Reveals Understanding

The sequence 3,245 -> 4,070 -> 6,308 is deliberate. The first number has all four places occupied. The next two contain internal zeros, which reveal whether the student understands placeholders or simply reads digits from left to right.

4. Explanation Is Part of the Mathematics

A correct answer does not by itself prove secure understanding. Ask the child to point, build, compare, and explain. Avoid supplying vocabulary too early; first listen to the child's idea, then refine the language.

Materials: Preparation and Use

Set up the table before the student arrives

MATERIALS

Print and Prepare

- Print student pages 1-8 single-sided.
- Print page 9 digit cards and pages 10-11 place-value pieces on cardstock, then cut on dashed lines.
- Put student page 3, the place-value mat, in a sheet protector if possible.
- Prepare a pencil, eraser, dry-erase marker, and two blank sheets for teacher modeling.
- Keep the answer pages out of the student's view.

What Each Material Does

Place-value mat	Organizes units by position	Put one type of piece in each labeled column. Never mix hundreds into the tens column.
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Value pieces	Represent actual quantities	A card marked 100 means one hundred, not one object to count as 1.
Digit cards	Represent standard notation	Place one digit below each column only after the quantity has been built.
Student packet	Records thinking	Use pp. 1-2 diagnostically; pp. 4-6 for practice; pp. 7-8 for transfer and assessment.
Blocks/disks	Optional stronger concrete model	Use in place of printed pieces. Insist on a consistent value for every shape or color.

Table Layout

- Center: place-value mat.
- Above the mat: piles of thousands, hundreds, tens, and ones, clearly separated.
- Below the mat: digit cards 0-9.
- Student's dominant-hand side: packet and pencil.
- Teacher's side: observation notes. Do not write on the student's paper during the diagnostic.

Material rule: When the student builds a number, require pieces to be placed in the matching column. When trading, physically remove one piece and replace it with 10 equal-value pieces before writing an equation.

Phase 1: Welcome and Diagnostic

Observe before teaching

STUDENT PAGES 1-2

Diagnostic	8-12 min	Packet pp. 1-2
Find out what the student already understands.	TIME	MATERIAL

Set the Tone

Teacher says	"Today I want to see how you think about numbers. Some questions may be easy and some may be unfamiliar. This is not a grade. If you are unsure, show any idea you have."
Why	This reduces performance pressure and makes the diagnostic more truthful.
Listen for	A willingness to attempt, draw, or explain.
Then do	Give neutral encouragement: 'Thank you for showing your thinking.' Do not say whether an answer is right yet.

How to Administer Page 1

- Ask the student to read 4,706; 3,090; and 8,215 aloud. Do not pronounce them first.
- For expanded form and comparison, say only: 'Show what you know.'
- For rounding, let the student draw any number line they understand.
- If the student asks what a word means, clarify ordinary language but do not teach the mathematics.

How to Administer Page 2

Teacher says	"Please solve these in your head as much as you can. Write a few marks or words so I can see the route your mind took. You do not need to use the standard vertical method."
Why	The goal is to reveal number relationships and flexibility, not speed.
Listen for	Compensation, making friendly numbers, decomposing tens and ones, or counting by ones.
Then do	After each problem ask only, 'How did you get that?' Record the strategy without correcting it.

Diagnostic Observation Codes

S	Secure	Accurate and explains using place value or an efficient relationship.
D	Developing	Mostly accurate but relies on prompts, counting, or imprecise language.
F	Fragile	Confuses places, ignores zeros, or cannot connect a digit to its value.
N/O	Not observed	No attempt or insufficient evidence; revisit later without treating it as failure.

Using the Diagnostic

Decide how much support to give in the lesson

TEACHER DECISION

What the Responses Mean

Reads 3,090 as 'three thousand ninety'	Understands zero placeholders in speech	Continue to 4,070 and 6,308.
Reads it as 'three thousand nine hundred'	Hundreds and tens are being confused	Return to the mat and point to every column.
Writes 6,384 as 6,000 + 300 + 80 + 4	Connects digit position to value	Ask for a nonstandard decomposition later.

Solves $399 + 58$ as $400 + 58 - 1$	Uses compensation flexibly	Invite multiple strategies, not extra routine work.
Counts $399 + 58$ by ones	Calculation is possible but inefficient	Do not reteach now; note for Day 4 mental-math work.

Choose the Lesson Path

If secure: Move briskly through the first model. Ask for alternate decompositions and justifications.
If developing: Teach the lesson as scripted. Keep pieces on the mat while recording symbols.
If fragile: Reduce the first number to 245. Establish hundreds, tens, and ones before adding thousands. Skip the hardest challenge if needed.

Transition to Teaching

Teacher says	"Thank you. I learned something about how you see numbers. Now we are going to investigate why the same digit can be worth very different amounts depending on where it is placed."
Why	Clearly closes assessment mode and opens collaborative learning mode.

Phase 2: Number Talk

Compose 1,000 in many ways

ORAL REASONING

Number Talk	5-8 min	Loose value pieces
Establish 10-to-1 trading before four-digit modeling.	TIME	MATERIAL

Launch

Teacher says	"Here is one thousand. How else could we make exactly the same amount using hundreds, tens, or ones?"
Why	Elicit equivalence: the appearance and number of pieces can change while total value stays fixed.
Listen for	'10 hundreds,' '100 tens,' or mixed decompositions.
Then do	After each answer ask, 'How could we prove it?' Build or sketch the proposed trade.

Use the Materials

- Place one 1,000 piece in the thousands column.

- Remove it and place ten 100 pieces in the hundreds column. Count by hundreds: 100, 200, ... 1,000.
- If pieces are limited, draw ten labeled hundred boxes rather than pretending fewer pieces are enough.
- Record beside the mat: $1,000 = 10 \text{ hundreds}$.
- Then ask how 10 hundreds could become tens. You may record $1,000 = 100 \text{ tens}$ without laying out all 100 pieces.

Productive Follow-Ups

Teacher says	"Could 9 hundreds equal 1,000? What is missing?"
Listen for	100 more, or 10 tens.

Teacher says	"If I use more pieces, does that always mean I have a greater number?"
Listen for	No; piece value matters.

Teacher says	"What stayed the same when we traded?"
Listen for	The total value.

Do not say: 'We add a zero' or 'Move it over.' Those shortcuts hide the quantity relationship. Say, 'Ten of this unit compose one of the next unit.'

Phase 3: Model 3,245

Connect quantity, position, and notation

STUDENT PAGE 3

Explicit Model	12-15 min	Mat, pieces, digit cards
Build one complete four-digit example.	TIME	MATERIAL

Step 1: Build the Quantity

Teacher says	"We are going to build three thousand two hundred forty-five. Before touching the pieces, what units do you expect we will need?"
Why	Prediction activates the structure before manipulation.
Listen for	Thousands, hundreds, tens, and ones.
Then do	Have the student place 3 thousand pieces, 2 hundred pieces, 4 ten pieces, and 5 one pieces in the matching columns.

Teacher says	"Point to the digit 2. What digit is it? What is it worth in this number? How can the pieces prove that?"
Listen for	'It is the digit 2, but it is worth 200 because there are two hundreds.'
Then do	If the child says only 'hundreds,' prompt: 'How many hundreds? What total value?'

Step 2: Add Digit Cards

- Place digit card 3 below the thousands column, 2 below hundreds, 4 below tens, and 5 below ones.
- Slide the cards together to form 3,245.
- Point from each card to its corresponding pieces.
- Do not place the digit cards first; quantity should lead notation in this first model.

Step 3: Record Four Forms

Teacher says	"We built one number. Now we will name the same number in several languages."
Why	The child should see forms as equivalent representations, not separate facts.
Then do	On student page 3 record: 3,245; $3,000 + 200 + 40 + 5$; three thousand two hundred forty-five; 3 thousands, 2 hundreds, 4 tens, 5 ones.

Check: Cover the pieces and ask the student to rebuild from 3,245. Then cover the numeral and ask the student to write it from the pieces.

Phase 4: Zeros and Placeholders

Teach 4,070 and 6,308

VARIATION

Placeholder Models	8-10 min	Mat + digit cards
Reveal whether zero is understood as an occupied place with no units.	TIME	MATERIAL

Model 4,070

Teacher says	"Build four thousand seventy. Which columns will contain pieces? Which column must remain empty?"
Listen for	4 thousands, 0 hundreds, 7 tens, 0 ones.
Then do	Leave the hundreds and ones columns visibly empty. Place zero digit cards below those columns.

Teacher says	"Why can we not write 47 for this model?"
Why	Make the placeholder function explicit.
Listen for	'47 would mean 4 tens and 7 ones. The zeros hold the thousands and tens in their correct positions.'
Then do	Compare 4,070 and 47 side-by-side with the mat.

Model 6,308

Teacher says	"Now build six thousand three hundred eight. Tell me what belongs in every column, including the empty one."
Listen for	6 thousands, 3 hundreds, 0 tens, 8 ones.
Then do	Ask the student to place the zero card in the tens position and read the completed number aloud.

Common Misconceptions

Student reads 4,070 as 'four thousand seven hundred.'	Point column by column: 'How many hundreds are shown? How many tens?' Rebuild rather than merely correcting the reading.
Student omits zero and writes 470.	Place 4 in the thousands column. Ask, 'If we remove this zero, where does the 4 move, and what would it be worth?'
Student says zero has no purpose.	Agree that it represents no pieces, then add: 'It still holds a place so the other digits keep their values.'
Student counts printed pieces as single objects.	Ask the child to read the label on each piece and count by its value: 100, 200, 300.

Phase 5: Trading Without Changing Value

Make regrouping conceptually visible

STUDENT PAGE 6

Trading	8-10 min	Mat + pieces
Show that a number can be decomposed in more than one equivalent way.	TIME	MATERIAL

Demonstrate the Trade

Teacher says	"Look at 3,245. I am going to remove one thousand piece. Have I changed the number yet?"
Listen for	'Yes, it is now 1,000 less.'
Then do	Confirm: 'Right now it has changed. To preserve the value, what must replace that thousand?'

Teacher says	"Replace the one thousand with ten hundreds. Count the hundreds now. Did the total number change? Prove it."
Why	Regrouping is an exchange of equal value, not a digit trick.
Listen for	12 hundreds total; $2,000 + 1,200 + 40 + 5$ still equals 3,245.
Then do	Record both decompositions directly under one another.

Write the Equivalence

- Usual form: 3 thousands + 2 hundreds + 4 tens + 5 ones.
- After trading: 2 thousands + 12 hundreds + 4 tens + 5 ones.
- Expanded values: $3,000 + 200 + 40 + 5 = 2,000 + 1,200 + 40 + 5$.
- Ask the student to circle the part that changed and underline the total that stayed equal.

Guided Example: 4,070

Teacher says	"Trade one thousand in 4,070 for hundreds. What is the new decomposition?"
Listen for	3 thousands + 10 hundreds + 7 tens.
Then do	If the student writes 3,107, return to units: 3 thousands, 10 hundreds, 7 tens is not read by concatenating the counts.

Why this matters later: This physical exchange is the foundation for regrouping in addition and subtraction. A child who understands the exchange will be less dependent on memorized carrying and borrowing rules.

Phase 6: Guided and Independent Practice

Move responsibility gradually to the student

STUDENT PAGES 4-6

Practice	12-15 min	Packet + mat
Shift from teacher model to independent representation.	TIME	MATERIAL

Use an I Do -> We Do -> You Do Sequence

I do	3,245	Teacher models briefly while student explains each placement.
We do	4,070 and 6,308	Student places pieces; teacher asks questions and records only after agreement.
You do	2,619; 5,040; 7,503	Student builds and records. Teacher observes and prompts minimally.

Prompt Ladder

First prompt: What do you notice? What could you try?

Second prompt: Which place should you examine first?

Third prompt: Point to the column for hundreds. What belongs there?

Last resort: Model a smaller related number, then return to the original.

What to Say During Page 5

Teacher says	"These are not three different numbers. They are three names for the same number. After completing a row, point across it and prove that every form has the same value."
Listen for	Connections between each digit and its expanded value.
Then do	For word form, have the student read aloud and point to each place. Do not penalize handwriting or spelling if the mathematical wording is clear.

What to Say During Page 6

Teacher says	"Your second decomposition must look different but have exactly the same total. Show the trade that makes it legal."
Listen for	A valid exchange of 1 unit for 10 units to the right.
Then do	Reject unexplained digit rearrangement. Ask the student to prove equality with pieces or expanded values.

Phase 7: Fourteen-Piece Challenge

Competition habit: organize possibilities

STUDENT PAGE 7

Challenge	7-10 min	Packet p. 7 + pieces
Search systematically and justify multiple solutions.	TIME	MATERIAL

Launch Without Giving a Method

Teacher says	"A piece may be worth 1,000, 100, 10, or 1. Build a four-digit number using exactly 14 pieces. There is more than one answer. Start anywhere, but be ready to prove your count."
Why	The problem develops organized search, place-value composition, and justification.
Listen for	The student checks that thousands + hundreds + tens + ones counts total 14.
Then do	Wait at least two minutes before suggesting a strategy.

If the Student Is Stuck

Teacher says	"A four-digit number needs at least how many thousand pieces?"
Listen for	At least 1.

Teacher says	"Suppose you choose 3 thousands. How many pieces remain for the other columns?"
Listen for	11.

Teacher says	"How could a table help you avoid repeating an answer?"
Then do	Use the six-column table on page 7.

Extend the Reasoning

Teacher says	"What is the greatest ordinary four-digit number possible with 14 pieces? What is the least? Convince me no better answer exists."
Listen for	Greatest 9,500; least 1,049, assuming each column is written with a standard digit 0-9.

Then do	Ask the student to explain why pieces should be moved toward the left for the greatest number and toward the right for the least.
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Competition habit: Praise organization and proof, not just finding an answer: 'Your table makes it clear you did not repeat a case.'

Phase 8: Exit Ticket and Feedback

Assess the central place-value idea

STUDENT PAGE 8

Exit Ticket	5-7 min	Packet p. 8
Decide whether the student understands tenfold place-value relationships.	TIME	MATERIAL

Give the Exit Ticket Independently

Teacher says	"Please answer these without my help. Use words, an equation, or a drawing to make your explanation convincing."
Why	The teacher needs independent evidence after instruction.
Then do	Do not prompt until the student has finished. Then ask one clarifying question if the writing is ambiguous.

Expected Explanation

Strong answer: The 5 in 5,204 is worth 5,000. The 5 in 52 is worth 50. The first 5 is two places farther left, and each move left multiplies value by 10. Therefore 5,000 is 100 times 50.

Respond to the Student

Secure	Correct value and tenfold explanation	Say: "You connected position to multiplication by 10. Tomorrow we can push that idea further."
Developing	Correct values but weak explanation	Say: "Your values are correct. Let us use the mat once more to show why the relationship is 100 times."
Fragile	Calls both digits 5 or confuses 500/5,000	Say: "The digit is the same, but its value changes. Tomorrow we will begin by rebuilding both numbers."

Close Positively

Teacher says	"Today you showed that a number can look different in pieces, words, and equations while keeping the same value. Tell me one idea from today that you could teach someone else."
Why	Retrieval and student teaching strengthen memory while ending on competence.

Answer Key and Observation Record

Use after the student has completed the work

REFERENCE

Diagnostic Answers

Read numbers

4,706: four thousand seven hundred six. 3,090: three thousand ninety. 8,215: eight thousand two hundred fifteen.

Expanded form

$6,384 = 6,000 + 300 + 80 + 4$.

Comparison

$4,099 < 4,900$.

Rounding

647 → 650 to nearest 10; 647 → 600 to nearest 100.

Mental math

$399 + 58 = 457$; $1,000 - 250 = 750$; $75 + 68 = 143$; $500 - 198 = 302$.

More than 3,794

1 more: 3,795. 10 more: 3,804. 100 more: 3,894.

Practice Answers

3,245	3 thousands, 2 hundreds, 4 tens, 5 ones
4,070	4 thousands, 0 hundreds, 7 tens, 0 ones
6,308	6 thousands, 3 hundreds, 0 tens, 8 ones
2,619	2 thousands, 6 hundreds, 1 ten, 9 ones
5,040	5 thousands, 0 hundreds, 4 tens, 0 ones
7,503	7 thousands, 5 hundreds, 0 tens, 3 ones

Trading and Challenge

3,245 after one thousand trade

2 thousands + 12 hundreds + 4 tens + 5 ones.

4,070 after one thousand trade

3 thousands + 10 hundreds + 7 tens.

8,462 alternate example

7 thousands + 14 hundreds + 6 tens + 2 ones.

Fourteen-piece examples

5,432 and 2,741 each use exactly 14 pieces.

Challenge extremes

Greatest: 9,500. Least four-digit number: 1,049.

Teacher Observation Record

Complete immediately after the lesson

PLANNING FOR DAY 2

Evidence Checklist

☐ Reads numbers containing internal zeros accurately.	☐ Names both the digit and its value.
☐ Matches pieces to the correct place-value column.	☐ Uses zero as a placeholder.
☐ Trades 1 unit for 10 units to the right without changing total value.	☐ Writes standard, expanded, word, and unit forms.
☐ Explains a tenfold relationship.	☐ Persists and organizes work during the challenge.

Record Specific Evidence

A strong statement or strategy the student used:

A misconception or hesitation to revisit:

Material that helped most:

Decision for Day 2

Secure: Proceed to expanded form and comparing numbers; include extension questions.

Developing: Proceed, but keep the mat and digit cards available.

Fragile: Begin Day 2 with a 10-minute small-number reteach using 245, 407, and 608.